SMARTBRIDGE

Let's Bridge the Gap

Date:	4 July 2026
Team ID:	PNT2022TMID124356
Project Title:	Credit Card Approval Prediction System
Maximum Marks:	5 Marks

Project Proposal (Proposed Solution)

SKILL WALLET — A SmartBridge Product

The proposal report aims to transform credit card approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Initialization and Planning Phase

Project Overview	
Objective	The primary objective is to revolutionize the credit card approval process by implementing advanced machine learning techniques, ensuring faster and more accurate assessments.
Scope	The project comprehensively assesses and enhances the credit card approval process, incorporating machine learning for a more robust and efficient system.
Problem Statement	
Description	Addressing inaccuracies and inefficiencies in the current credit card approval system adversely affects operational efficiency and customer satisfaction.
Impact	Solving these issues will result in improved operational efficiency, reduced financial risk, and an overall enhancement in the card issuance process, contributing to higher customer satisfaction and organizational success.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict customer creditworthiness, creating a dynamic, data-driven, and adaptable credit card approval system.
Key Features	• Implementation of a machine learning-based credit assessment model. • Real-time decision-making for quicker credit card approvals. • Continuous learning workflows to adapt to evolving consumer financial landscapes.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask / Streamlit
Libraries	Additional data science and ML libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, xgboost
Development Environment	IDE	Jupyter Notebook, PyCharm / VS Code
Data		
Application Data	Source, size, format (Credit Card Application dataset)	Kaggle dataset, 614 rows, .csv
Credit Data	Source, size, format (Credit Card Record dataset)	UCI Machine Learning Repository, 690 rows, .csv